PAIHENG XU

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RESEARCH INTEREST

Natural Language Processing, Computational Social Science, AI for Education, Causal Inference

EDUCATION

University of Maryland, College Park, USA *2021 - present*
Ph.D. in Computer Science
Thesis: From Measurement to Discovery: Comparing Text in Computational Social Science.

Johns Hopkins University, Baltimore, USA *2018 - 2020*
M.S.E in Computer Science

Southwest University, Chongqing, China *2014 - 2018*
B.E in Computer Science

RESEARCH EXPERIENCE

UNIVERSITY OF MARYLAND, COLLEGE PARK — College Park, MD
Research Assistant – Advisor: Wei Ai & Jing Liu *Sep. 2021 - present*

- **Hypothesis generation** for text analysis using **Sparse Autoencoder** and causality-inspired methods.
- NLP-based **construct measurement** of teaching quality from classroom transcripts.
- Developing AI teacher-coaching benchmarks and **contextualized rubrics** for open-ended evaluation.
- **Bias** and **spurious correlation** mitigation for text classification and **LLM alignment**.

Research Assistant – Advisor: Louiqa Raschid & Vanessa Frias-Martinez *Jan. 2023 - May 2025*

- Model correlations between geo-co-location and social media engagement using **statistical** & **LLM**-based methods.

JOHNS HOPKINS UNIVERSITY — Baltimore, MD
Full-time Research Developer – Advisor: Mark Dredze *Jun. 2020 - Aug. 2021*

- Develop Twitter-based social mobility indices to study demographic differences in social distancing behavior. [link]
- Apply embedding- and transformer-based models to discover e-cig brands and track sentiment shifts on social media.

Research Assistant – Mentor: Zach Wood-Doughty *Nov. 2018 - May 2020*

- Demographic inference on Twitter with distant supervision (Package); study linguistic variation across racial groups.

INDUSTRY EXPERIENCE

ADOBE RESEARCH — San Jose, CA
Research Scientist Intern – Mentor: Haoliang Wang *May 2025 - Aug. 2025*

- Design pattern discovery from user histories using VLMs to generate personalized design suggestions.
- Personalization evaluation using targeted image edits to assess alignment with user-specific design patterns.

ADOBE RESEARCH — San Jose, CA
Research Scientist Intern – Mentor: Vishy Swaminathan, Gang Wu, & Xiang Chen *May 2024 - Nov. 2024*

- **Skill discovery** for software scripting through code generation & offline simulations using **LLM-based agents**.

ADOBE RESEARCH — Remote
Research Scientist Intern – Mentor: Tong Yu & Haoliang Wang *May 2022 - Aug. 2022*

- Knowledge extraction from service outage reports for automated remediation.
- Adaptive question generation for extractive question answering using in-context learning.

SELECTED PUBLICATIONS

* denotes equal contribution; † denotes mentoring contribution

Paiheng Xu, Jing Liu, Wei Ai, Conditional Hypothesis Generation for LLM-Based Text Analysis with Researcher-Specified Covariates. *arXiv:2606.03029, 2026*

Jiseung Yoo, **Paiheng Xu**†, Jing Liu, More Than Words: A Multimodal AI Analysis of Teacher Praise in Online Tutoring. *Under review.*

Nishant Balepur, **Paiheng Xu**, Wei Ai, Eunsol Choi, Rachel Rudinger, Jordan Lee Boyd-Graber, Check The Scoreboard: An Analysis of Scoring Schemes on Multiple-Choice Evaluation. *EMNLP 2026.*

Paiheng Xu, Louiqa Raschid, Vanessa Frias-Martinez, Does Geo-co-location Matter? A Case Study of Public Health Conversations during COVID-19. *ICWSM 2026.*

Paiheng Xu, Gang Wu, Xiang Chen, Tong Yu, ..., Tianyi Zhou, Wei Ai, Vishy Swaminathan, Skill Discovery for Software Scripting Automation via Offline Simulations with LLMs. *Findings of EACL*, 2026.

Paiheng Xu*, Xiaoyu Liu*, Junda Wu, ..., Tong Yu, Julian McAuley, Wei Ai, Furong Huang, Large Language Models and Causal Inference in Collaboration: A Comprehensive Survey. *Findings of NAACL 2025.*

Paiheng Xu*, Yuhang Zhou*, Bang An, Wei Ai, Furong Huang, GFairHint: Improving Individual Fairness for Graph Neural Networks via Fairness Hint. *Transactions on Knowledge Discovery from Data, 2025*

Paiheng Xu, Jing Liu, Nathan Jones, Julie Cohen, Wei Ai, The Promises and Pitfalls of Using Language Models to Measure Instruction Quality in Education. *NAACL 2024.*

Yuhang Zhou, **Paiheng Xu**, Xiaoyu Liu, Bang An, Wei Ai, Furong Huang, Explore Spurious Correlations at the Concept Level in Language Models for Text Classification. *ACL 2024.*

Paiheng Xu, David A. Broniatowski, Mark Dredze, Twitter Social Mobility Data Reveal Demographic Variations in Social Distancing Practices During the COVID-19 Pandemic. *Scientific Reports*, 2024.

Yuhang Zhou*, Jing Zhu*, **Paiheng Xu**, ..., Danai Koutra, Wei Ai, Furong Huang, Multi-Stage Balanced Distillation: Addressing Long-Tail Challenges in Sequence-Level Knowledge Distillation. Findings of EMNLP 2024.

Paiheng Xu*, Zach Wood-Doughty*, Xiao Liu, Mark Dredze, Using Noisy Self-Reports to Predict Twitter User Demographics. *SocialNLP@NAACL*, 2021.

Paiheng Xu, Mark Dredze, David A. Broniatowski, The Twitter Social Mobility Index: Measuring Social Distancing Practices from Geolocated Tweets. *Journal of Medical Internet Research*, 2020.

Paiheng Xu, Likang Yin, Zhongtao Yue, Tao Zhou, On Predictability of Time Series. *Physica A: Statistical Mechanics, its Applications* 523, 345-351 (2019).

TECHNICAL SKILLS

Languages	Python, MATLAB, R, C/C++, C#, LaTeX, Bash
Tools	Transformers, PyTorch, TRL, HF Agents, Scikit-learn, NLTK, Statsmodels, NetworkX

HONOR & AWARDS

Dean's Fellowship, University of Maryland
National Scholarship, Ministry of Education, China
Outstanding Graduates, Southwest University

TEACHING & SERVICE

Teaching Assistant: Introduction to Artificial Intelligence, Fall 2022
Program Committee: ICWSM 2022, 2025-26
Reviewer: ARR 2025-26; KDD 2025; TheWebConf 2025; TPAMI
Sub-Reviewer: ACL 2021; KDD 2022; SIGIR 2022-2025